Swap Components and Affine Rank in Knapsack Facet Recognition

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Research checkpoint, October 13, 2026

Abstract

We record a structural route toward the fixed-parameter question posed by Chen and Zhu. For a valid supporting inequality with positive coefficients, let C be the number of connected components of its feasible equal-coefficient swap graph. We show that its tight-face dimension can be computed using fixed-dimensional integer programming in dimension $2C$, without counting integer points. This gives an algorithm parameterized by C, not a solution parameterized solely by the number K of distinct coefficients. We also give an exact six-variable facet on which minimum-cost parity representatives, even together with every feasible swap, miss one rank. These are intermediate findings; the original FPT question remains open in this checkpoint.

1 Problem and known work

Let $a \in \mathbb{Z}^n_{>0}$, $b \in \mathbb{Z}_{>0}$, and $\max_i a_i \le b$. Define

$$P = \operatorname{conv}\{x \in \{0,1\}^n : a^\top x \le b\}.$$

The polytope has dimension n. Given $c \in \mathbb{Z}^n_{\ge 0}$ and $d \in \mathbb{Z}_{\ge 0}$, the task is to recognize whether $c^\top x \le d$ defines a facet. The parameter K is the number of distinct positive entries of c; L denotes the binary input length. Chen and Zhu prove general DP-completeness, as well as an $n^{K+O(1)}$ recognition and face-dimension algorithm [1]. Their open question asks for fixed-parameter tractability in K.

An existing public progress note [2] disproves a two-anchor spanning claim, proves a valid swap-completion lemma, and reduces the affine-rank calculation to integer-point counting in dimension $2K$. Fixed-dimensional counting alone does not establish a uniform $f(K)\,\mathrm{poly}(n, L)$ bound. We use its swap idea but replace counting by a different parameter, namely the number of swap components. No claim is made that this new parameter is already bounded by K.

2 The swap graph

First assume $c_i > 0$ for every i, and that the inequality is valid and supporting. Write

$$T = \{x \in \{0,1\}^n : a^\top x \le b,\ c^\top x = d\}.$$

Order each equal-coefficient class by nondecreasing weight. Join two items $r < s$ in the same class when there is $x \in T$ with $x_r = 0$ and $x_s = 1$. Such a point can swap s for r and remain in T. Let $B_1, \dots, B_C$ be the connected components of this graph, and define the linear map

$$(L_B x)_\ell = \sum_{i \in B_\ell} x_i \qquad (1 \le \ell \le C).$$

Here L_B is a map, whereas L without a subscript remains the input length. Let $D = \operatorname{span}(T - x^0)$ for any fixed $x^0 \in T$.

The following observation makes it sufficient to test adjacent pairs within each coefficient class.

Lemma 1. *If $r < s$ is a feasible swap pair, every pair $u < v$ with $r \le u < v \le s$ is feasible. Consequently, each graph component is an interval, and the component partition is determined by the adjacent feasible swap pairs.*

Proof. Take a witness $x_r = 0, x_s = 1$. If $x_u = 0, x_v = 1$, nothing is needed. If both are zero, replace s by v. If both are one, replace u by r. In the remaining case $x_u = 1, x_v = 0$, replace both u, s by r, v. In that case all four indices are distinct, and $a_r + a_v \le a_u + a_s$. Each replacement preserves the coefficient sum and does not increase the weight. Thus it yields a tight feasible witness with $x_u = 0, x_v = 1$. Any path connecting indices on opposite sides of an adjacent cut contains an edge crossing that cut; the first assertion makes that adjacent pair feasible. This proves the component statement. □

The dimension identity follows from the kernel of the component-count map.

Lemma 2. *Let $U = L_B(T) \subseteq \mathbb{Z}^C$. Then*

$$\dim \operatorname{aff}(T) = n - C + \dim \operatorname{aff}(U). \tag{1}$$

Proof. Each graph edge contributes a vector $e_r - e_s \in D$. In a connected component these edge vectors span every vector supported on that component whose coordinates sum to zero. Summing over components gives $\ker L_B \subseteq D$ and $\dim \ker L_B = n - C$. Moreover $L_B(D) = \operatorname{span}(U - L_B x^0)$. Apply rank-nullity to the restriction of L_B to D. □

3 A counting-free algorithm with a larger parameter

For component B_ℓ, put $m_\ell = |B_\ell|$, let η_ℓ be its common coefficient, sort its weights $w_{\ell 1} \le \cdots \le w_{\ell m_\ell}$, and put

$$G_\ell(t) = \sum_{j=1}^{t} w_{\ell j} \qquad (0 \le t \le m_\ell).$$

Choosing the cheapest items within every component shows exactly that

$$U = \left\{ u \in \mathbb{Z}^C : 0 \le u_\ell \le m_\ell, \quad \eta^\top u = d, \quad \sum_{\ell=1}^{C} G_\ell(u_\ell) \le b \right\}. \tag{2}$$

No relaxation of the feasible set occurs in this equality: every feasible count vector has an actual binary prefix realization.

Theorem 3. *For positive c, validity, support, and the swap components can be determined in $f(K)\,\mathrm{poly}(n, L)$ time. After the components are known, the tight-face dimension can be determined in $g(C)\,\mathrm{poly}(n, L)$ time. In particular, facet recognition is fixed-parameter tractable in C on valid supporting instances.*

Proof. In the original K coefficient classes, minimum cost for any prescribed integer count is a convex prefix-sum function. Introduce one integer count and one integer epigraph variable per class. The epigraph of a prefix-sum function F on $\{0, \dots, m\}$ is represented by the inequalities

$$y \ge F(q) + w_{q+1}(z - q), \qquad 0 \le q < m.$$

Thus validity is tested by integer feasibility with candidate value at least $d+1$, and support by feasibility with value d. There are $2K$ variables, $O(n+K)$ inequalities, and polynomial input bit length. Fixed-dimensional integer programming gives the required FPT bound; we use the uniform parameterized formulation in [4, Theorem 1], building on Lenstra [3] and subsequent improvements. The convex-prefix epigraph technique is an existing tool, also covered by [4, Theorem 2], rather than a novelty claim here.

For a proposed adjacent swap $r < s$, force r out and s in. In that class, for count $t \in \{1, \dots, m-1\}$ the minimum cost is a_s plus the sum of the $t-1$ cheapest weights after deleting r, s. This is again convex in t. Use the same $2K$-variable formulation, restricting the count to that domain. There are at most $n-K$ adjacent tests. Lemma 1 gives the full component partition.

Represent (2) with C count variables and C epigraph variables, using at most $O(n+C)$ inequalities. Find one feasible u^0. To find its affine hull, maintain a rational space W spanned by the count differences found so far. For an integer-scaled basis q of $W^\perp$, test the two systems obtained by adjoining

$$q^\top(u-u^0) \geq 1 \quad \text{or} \quad q^\top(u-u^0) \leq -1.$$

If either system has a solution, its difference from u^0 increases the dimension of W. If every test fails, every feasible count lies in u^0+W, so W is the exact difference space. At most C dimension increases and $O(C^2)$ feasibility calls are needed. Rational linear algebra has polynomial bit complexity since counts are bounded by n. Each integer feasibility call has $2C$ variables. Finally use (1); facetness is equivalent to $\dim \operatorname{aff}(U) = C-1$. □

If zero coefficients occur, the reduction in Chen and Zhu's Proposition 2 can be used before this theorem. Check that a tight point exists containing the heaviest zero-coefficient item; a cheapest-positive-count feasibility test handles this in parameter K. If the check passes, facetness is equivalent to facetness on the positive-coefficient support. The identically zero candidate does not define a proper facet. This preprocessing does not assert that zero coefficients are positive or silently change the parameter.

The remaining issue is substantial: C is computed from the instance, and may be as large as n on general supporting faces. A bound $C \leq h(K)$ *for facet instances* would suffice to finish the original problem: after computing C, reject instances violating that necessary bound and apply the theorem otherwise. No such bound is proved here. Nor do the experiments establish one.

4 An exact obstruction to parity representatives

A tempting alternative keeps, for each parity class z mod 2, one minimum-weight tight count profile, and adds every feasible within-class swap. This need not preserve even the rank of a genuine facet.

Set

$$a = (5,8,3,3,3,3), \quad b = 13, \quad c = (2,2,1,1,1,1), \quad d = 4.$$

There are two coefficient classes, $A = \{1,2\}$ and $B = \{3,4,5,6\}$. The complete tight set is

$$T = \{(1,1,0,0,0,0),\ (0,0,1,1,1,1)\} \ \cup\ \{(1,0,y) : y \in \{0,1\}^4,\ \textstyle\sum_i y_i = 2\}. \tag{3}$$

Indeed, with zero, one, or two A-items chosen, coefficient value greater than four would require respectively more than four B-items, at least three B-items (weight at least $5+9=14$), or at least one B-item (weight at least $5+8+3=16$). None is feasible. At equality, choosing item 2 alone with two B-items has weight $8+6=14>13$, whereas item 1 with two B-items has weight 11. This proves both validity and (3).

The six vectors $y \in \{0,1\}^4$ of coordinate sum two have affine dimension three: their differences span the zero-sum subspace of $\mathbb{R}^4$. Their lifted points $(1,0,y)$ still have affine dimension three. Adding $(0,0,1,1,1,1)$ increases the dimension by one, because its first coordinate is zero. Adding $(1,1,0,0,0,0)$ increases it again, because its second coordinate is one. The tight face therefore has dimension five and is a facet of the six-dimensional knapsack polytope.

Count profile (z_A, z_B)	Weight	Parity	Minimum-cost representative?
$(0,4)$	12	$(0,0)$	Yes
$(1,2)$	11	$(1,0)$	Yes
$(2,0)$	13	$(0,0)$	No

Table 1: All tight profiles in the six-variable counterexample.

All four B-items belong to one swap component. There is no feasible swap between the two A-items, since the needed tight witness would select item 2 alone and two B-items. Thus $C = 3 > K = 2$. Every selected parity representative has $x_2 = 0$, as does every feasible swap direction. Their augmented span has rank five and cannot contain $(1,1,0,0,0,0)$, while the full augmented span has rank six. This also shows why equal-coefficient classes cannot simply be treated as connected swap components.

The exact verifier enumerates all $2^6 = 64$ binary vectors, confirms the full tight set, and checks a nonzero augmented 6×6 minor. The larger 21-variable example that led to this simplification is retained in the evidence package. Neither example is a counterexample to fixed-parameter tractability.

5 Status and reproducibility

The results above isolate a structural question; their novelty is not certified. The evidence package provides exact arithmetic verifiers, complete inputs, a facet basis, and a nonzero determinant. Floating-point exploratory ranks are used only for discovery. There has been no independent external review.

References

[1] R. Chen and H. Zhu, *The Complexity of Recognizing Facets for the Knapsack Polytope*, Mathematics of Operations Research, published online November 17, 2025, doi:10.1287/moor.2024.0481. Author manuscript: arXiv:2211.03311v3, October 20, 2025.

[2] Author not identified, *A Rank-Counting Reduction for Knapsack Facet Recognition with Few Candidate Coefficients: Corrected progress and an obstruction to two-anchor spanning*, June 25, 2026. Public progress note in `AI-in-Math-OR/open-problems-in-or`, path `data/llm_math_export/solution_progress/W4416295153_p0/partial_progress/34.pdf`. Retrieved October 13, 2026; exact source hash retained in the evidence manifest.

[3] H. W. Lenstra, Jr., *Integer programming with a fixed number of variables*, Mathematics of Operations Research 8(4) (1983), 538–548, doi:10.1287/moor.8.4.538.

[4] R. Bredereck, P. Faliszewski, R. Niedermeier, P. Skowron, and N. Talmon, *Mixed Integer Programming with Convex/Concave Constraints: Fixed-Parameter Tractability and Applications to Multicovering and Voting*, arXiv:1709.02850v2, November 20, 2017.